

TEST-08-REGULATORY-COMPLIANCE-MATRIX

TEST-08: REGULATORY COMPLIANCE TESTING MATRIX

Carrington Storm Motors — Project AEGIS

Complete Standards-to-Product Mapping — UL • FCC • NIST • MIL-STD • IEC • ISO • ASTM • AASHTO • ABS • DNV

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Classification: CONFIDENTIAL — Accredited Investor Eyes Only
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Reference: ROAD-05 (Certification & Regulatory Pathway) — this matrix operationalizes ROAD-05 into specific test requirements per product

COLOR CODING

Color	Meaning
GREEN	Standard applicable; testing complete; certificate issued
AMBER	Standard applicable; testing in progress
RED	Standard applicable; testing not yet initiated — URGENT (blocking regulatory approval or customer requirement)
◦ WHITE	Standard applicable; testing scheduled in future quarter
GRAY	Standard not applicable to this product

SECTION 1: ELECTRICAL SAFETY (UL)

Product / Product Group	UL 94 (Flammability)	UL 723/E84 (Surface Burning)	UL 746C (Weathering)	UL 263/E119 (Fire Resist)	UL Product-Specific
Aegis-C Wall Panel	AMBER	AMBER	AMBER	RED (Q1 2027)	—
BFRP Rebar	GRAY	GRAY	GRAY	AMBER (with geopolymers system)	—
Safe-Wave Microwave	◦ WHITE (Q3 2027)	GRAY	GRAY	GRAY	UL 923: ◦ Q3 2027
Stellar-Chill Refrigerator	GRAY	GRAY	GRAY	GRAY	UL 250: ◦ Q3 2027
Carrington-Proof Oven	◦ WHITE (Q3 2027)	GRAY	GRAY	GRAY	UL 858: ◦ Q3 2027
Ceramic Water Heater	GRAY	GRAY	GRAY	GRAY	UL 174: ◦ Q3 2027
Inductive-Proof Wiring (PMMA POF)	◦ WHITE (Q1 2027)	GRAY	GRAY	GRAY	UL 83 (custom OoI): ◦ Q2 2027
Non-Conductive	◦ WHITE (Q2)	GRAY	GRAY	GRAY	UL 1004: ◦ Q3

Motor	2027)				2027
Aegis Home Energy Panel	GRAY	GRAY	GRAY	GRAY	UL 508A: ◦ Q1 2028
Phantom MK-1 Battery	GRAY	GRAY	GRAY	GRAY	UL 1973: ◦ Q3 2028
Phantom MK-1 Complete Robot	GRAY	GRAY	GRAY	GRAY	UL 1740: ◦ Q1 2029
Ascension Drone Battery	GRAY	GRAY	GRAY	GRAY	UL 1642: ◦ Q3 2028
Protonic Transceiver	GRAY	GRAY	GRAY	GRAY	UL 62368-1: ◦ Q2 2029
Aegis Embark Products (8)	GRAY	GRAY	GRAY	GRAY	UL 2089 (modified): ◦ Q3 2028

SECTION 2: COMMUNICATIONS (FCC)

Product	FCC Part 15.247 (Intentional Radiator)	FCC Part 15, Class A/B (Unintentional)	FCC Petition / Exemption	Status
LoRa Mesh Radio (EMF Detector, FEATHER)	RED — Q1 2027 scheduled	GRAY	GRAY	FEATHER module uses pre-certified Semtech SX1276 — integration testing Q1 2027
Protonic Transceiver	GRAY	GRAY	AMBER — Petition filed Q4 2026 (target); ruling expected Q4 2027	Exemption based on “no RF radiation” physics argument
PMMA POF Data Network	GRAY	◦ WHITE — Q2 2027	GRAY	Minimal RF from optoelectronic transceivers
Phantom MK-1 Onboard Electronics	GRAY	◦ WHITE — Q4 2027	GRAY	Industrial unintentional radiator
Ascension Drone Telemetry	RED — Q2 2027 (Part 15.247 or Part 87)	GRAY	GRAY	May require FAA coordination if Part 87

SECTION 3: ELECTROMAGNETIC RESILIENCE (NIST / MIL-STD)

Product	MIL-STD-188-125 (HEMP)	MIL-STD-461 (EMI)	MIL-STD-810 (Environmental)	MIL-STD-882 (Safety)	NIST SP 800-53 (Cyber)
Phantom MK-1 Military Variant	RED — 2029 (White Sands)	◦ WHITE — 2028	◦ WHITE — 2028	◦ WHITE — 2028	◦ WHITE — 2028
Phantom MK-1 Civilian	GRAY (not pursuing)	◦ WHITE — 2027 (reference data)	◦ WHITE — 2027 (reference data)	◦ WHITE — 2028	GRAY
Protonic Network (Military)	GRAY (protonic immune by physics)	GRAY	◦ WHITE — 2028	◦ WHITE — 2029	◦ WHITE — 2029
Aegis Home (Military Family Housing)	GRAY (not pursuing)	GRAY	GRAY	GRAY	GRAY
Infrastructure Products (DoD Bases)	RED — 2029	◦ WHITE — 2028	◦ WHITE — 2028	◦ WHITE — 2028	◦ WHITE — 2028
Ascension Drone Military Variant	GRAY	◦ WHITE — 2029	◦ WHITE — 2029	◦ WHITE — 2029	◦ WHITE — 2029

SECTION 4: INTERNATIONAL ELECTROTECHNICAL (IEC)

Product Group	IEC 61000-4-2 (ESD)	IEC 61000-4-8 (Mag Field) — CRITICAL	IEC 61000-4-3 (RF Immunity)	IEC 60243 (Dielectric)	IEC 62368-1 (Safety)
Aegis Home Electronics	◦ Q2 2027	RED — Q3 2027	◦ Q2 2027	◦ Q2 2027	◦ Q3 2028
Aegis Embark Products	◦ Q3 2027	RED — Q4 2027	◦ Q3 2027	◦ Q2 2027	◦ Q1 2029
Phantom MK-1	◦ Q4 2027	RED — Q1 2028	◦ Q4 2027	◦ Q3 2027	◦ Q3 2029
Protonic Transceiver	GRAY (protonic immune)	AMBER — special test at 1,000 A/m	GRAY	◦ Q3 2027	◦ Q2 2029
Infrastructure Electronics	◦ Q3 2028	RED — Q4 2028	◦ Q3 2028	◦ Q2 2028	◦ Q4 2029
Charlemagne Vessel Systems	◦ Q1 2028	RED — Q2 2028	◦ Q1 2028	◦ Q4 2027	◦ Q2 2030

IEC 61000-4-8 (Power Frequency Magnetic Field Immunity) is flagged RED for all products because this is THE certification that validates Carrington hardening. It must be prioritized.

SECTION 5: QUALITY MANAGEMENT (ISO)

Standard	Scope	Timeline	Cost	Status
ISO 9001:2015	CSM Enterprise — Quality Management	Q1 2028	\$85K	AMBER — Gap analysis complete; QMS documentation in draft
ISO 14001:2015	Environmental Management	Q4 2028	\$65K	◦ WHITE
ISO 45001:2018	Occupational Health & Safety	Q4 2028	\$55K	◦ WHITE
ISO 27001:2022	Information Security (LandOLil cloud)	Q3 2027	\$48K	◦ WHITE
ISO 17025:2017	Testing Lab Competence	Q4 2029	\$120K	◦ WHITE
IATF 16949:2016	Automotive Quality (Aegis Embark)	Q2 2029	\$95K	◦ WHITE
AS9100D	Aerospace Quality (Ascension drone)	Q2 2029	\$85K	◦ WHITE

SECTION 6: BUILDING CODE (ICC / IAPMO / International)

Product	ICC-ES AC04 (Panels)	ICC-ES AC454 (BFRP Rebar)	ICC-ES AC470 (Geopolymer)	IAPMO (Plumbing)	California Title 24	EU ETA	Status Summary
Aegis-C Wall Panel	RED — Submit Q1 2027	GRAY	GRAY	GRAY	◦ Q3 2027	◦ Q1 2028	CRITICAL PATH
BFRP Rebar	GRAY	RED — Submit Q1 2027	GRAY	GRAY	◦ Q3 2027	◦ Q1 2028	CRITICAL PATH
Geopolymer	GRAY	GRAY	RED — Submit Q1	GRAY	◦ Q2 2027	◦ Q1 2028	CRITICAL

Concrete			2027			PATH
GFRP Pipe System	GRAY	GRAY	GRAY	RED — Q3 2027	GRAY	◦ Q2 2028
Complete Aegis Home	◦ Q4 2027 (AC481)	GRAY	GRAY	GRAY	◦ Q3 2027	◦ Q2 2028

SECTION 7: MARITIME CLASSIFICATION (ABS / DNV / LR)

Product / System	ABS (Steel Vessel Rules)	DNV (Class Rules)	Lloyd's Register	IMO SOLAS	IMO FTP Code (Fire)	Status
Charlemagne Hull Coating (MXene)	RED — Q2 2028	RED — Q2 2028	RED — Q3 2028	◦ Q3 2028	AMBER — component testing in progress	New tech qualification required
BFRP Structural Elements	RED — Q2 2028	RED — Q2 2028	RED — Q3 2028	◦ Q3 2028	AMBER	Alternative material approval
PMMA POF Data Network	RED — Q3 2028	RED — Q3 2028	RED — Q4 2028	◦ Q4 2028	RED — cable fire testing	
LiFePO ₄ Battery System	RED — Q3 2028	RED — Q3 2028	RED — Q4 2028	◦ Q4 2028	◦ Q1 2029	Battery propulsion guide
FOG Navigation	◦ Q1 2029	◦ Q1 2029	◦ Q2 2029	◦ Q4 2028	GRAY	Type approval
Dielectric Fire Suppression	RED — Q4 2028	RED — Q4 2028	RED — Q1 2029	RED — Q1 2029	◦ Q1 2029	Critical safety system
Protonic Emergency Comms	◦ Q2 2029	◦ Q2 2029	◦ Q3 2029	◦ Q2 2028 (life-saving)	GRAY	GMDSS supplementary

SECTION 8: AUTOMOTIVE (IATF / AEC / FMVSS)

Aegis Embark Product	IATF 16949 (QMS)	AEC-Q200 (Components)	FMVSS Applicable	Status
Neural-Grip MRE Steering Wrap	RED — Q2 2028	RED — Q2 2028	FMVSS 124: RED — Q3 2028	
Cervical-Guard Headrest	RED — Q2 2028	RED — Q2 2028	FMVSS 202a: RED — Q3 2028	
Mag-Float Seat Rail	RED — Q2 2028	RED — Q2 2028	FMVSS 207: RED — Q3 2028	
Thorax-Calm Seatbelt	RED — Q2 2028	RED — Q2 2028	FMVSS 209: RED — Q3 2028	
Stellar-Tint Window Film	RED — Q2 2028	RED — Q2 2028	FMVSS 205: RED — Q2 2028	
All 8 Products (Aftermarket)	GRAY	GRAY	NHTSA “Make Inoperative” (49 USC 30122): ◦ Q1 2028 — Letter of Interpretation	

SECTION 9: MATERIALS (ASTM)

Material	ASTM Standards Applicable	Status
MXene Coating	D257, D4935, D3359, B117, G154	GREEN — baseline testing complete; durability in progress

BFRP Composite	D3039, D7264, D2344, D570	GREEN — baseline complete
BFRP Rebar	D7205, D7914, D7957	AMBER — D7914 long-term creep in progress (6-month test)
Geopolymer Concrete	C39, C496, C1202, C1760	GREEN — baseline complete
Si ₃ N ₄ Ceramic Bearings	F2094	GREEN — specification verified
Polyimide-Silica Aerogel	C518, C165	GREEN — baseline complete
YInMn Blue Coating	G154, B117, D3359	AMBER — UV and salt spray in progress
PMMA POF Cable	D6577 (adapted), IEC 60794 series	AMBER — fiber characterization in progress
MRF-140CG Fluid	D2196 (adapted)	GREEN — rheological characterization complete

SECTION 10: PRIORITY BY URGENCY — TOP 10 ACTIONS

#	Product	Standard	Urgency	Target Date	Owner
1	Aegis-C Wall Panel	ICC-ES AC04	CRITICAL — blocks all Aegis Home commercial sales	Submit Q1 2027	Nyx + Chester
2	BFRP Rebar	ICC-ES AC454	CRITICAL — blocks structural use	Submit Q1 2027	Nyx + Chester
3	Geopolymer Concrete	ICC-ES AC470	CRITICAL — blocks foundation use	Submit Q1 2027	Nyx + Chester
4	All Electronic Products	IEC 61000-4-8	CRITICAL — core Carrington immunity certification	Q3 2027 (staggered by product)	Zirconia + all Agents
5	Protonic Transceiver	FCC Petition	HIGH — regulatory uncertainty	File Q4 2026	Vance + FCC counsel
6	Charlemagne Hull Coating	ABS/DNV NTA	CRITICAL — blocks vessel retrofit	Q2 2028	Vaun + class surveyors
7	Aegis Embark (8)	AEC-Q200	CRITICAL — blocks automotive supply	Q2 2028	All Embark Agents
8	Aegis Embark (8)	IATF 16949	CRITICAL — prerequisite for AEC-Q200 and OEM supply	Q2 2028	Nash (lead) + Chester
9	Aegis Home Wall Panel	UL 94/723/746C	HIGH — prerequisite for ICC-ES and builder acceptance	Results Q2 2027	Nyx + Chester
10	ISO 9001:2015	Enterprise QMS	HIGH — prerequisite for IATF, AS9100, ISO 17025	Q1 2028	Chester

REGULATORY COMPLIANCE BUDGET BY YEAR (from ROAD-05)

Year	Total Compliance Cost
2026 (Q3-Q4)	\$588K

2027	\$1.42M
2028	\$2.08M
2029	\$798K
2030	\$216K
5-Year Total \$5.10M	

“This matrix is the heartbeat of regulatory compliance. Every red cell is a blocker between a CSM product and its market. Every green cell is evidence that the Dielectric Citadel meets the standards of the civilization it protects. The priority is clear: building code approval unlocks Aegis Home; maritime classification unlocks Charlemagne; automotive certification unlocks Aegis Embark; IEC 61000-4-8 immunity unlocks the claim of Carrington hardening across all products. Clear the red cells. Earn the green cells. Ship the products.” — Chester, Director of Quality Assurance

Document generated 2026-07-15. Updated quarterly. Next update: 2026-10-01 (Q4 2026 gate review). All RAG statuses to be re-evaluated based on progress against ROAD-01 and ROAD-10 milestones. New standards may be added as regulatory landscape evolves (e.g., potential NERC GIC reliability standard, EU Cyber Resilience Act applicability to Protonic/IoT products).